# Group Project: Interactive ($C_p$ *vs.* T) Database for Engineering Materials

## Objective

Develop an interactive computational platform for plotting and analysing the variation of specific heat capacity at constant pressure ($C_p$), with temperature, (T), for a minimum of 200 different materials. The assignment should demonstrate how thermodynamic data can be converted into a visually appealing and user-friendly materials-property database.

## Instructions

1. This is a group assignment to be completed in teams of two students, except for one group, which consists of three students
2. The groups have already been formed and are provided in the attached Excel sheet.
3. Students are not permitted to change, exchange, or modify their group members.
4. Both students must contribute substantially to the development and presentation of the project.
5. Marks will be based on individual contribution to the project.
6. Use of AI tools will carry **negative or zero** marks

## Technical Requirements

Students may use either MATLAB, or Python, whoever is convenient.

The developed platform should calculate and display ($C_p$) as a function of temperature using equations and coefficients obtained from reliable thermodynamic tables or materials-property databases. The database must contain at least **200 materials**, covering different classes such as:

- Metals and alloys
- Ceramics
- Semiconductors
- Polymers
- Glasses
- Refractories
- Composite materials
- Other technologically relevant materials

Data may be collected from reliable sources such as thermodynamic handbooks, journal articles, NIST databases, the Materials Project, or other recognised materials-property databases. Every data source must be properly cited. **List of database:** NIST Chemistry WebBook, NIST JANAF Thermochemical Tables, Materials Project, AFLOW Database, Open Quantum Materials Database (OQMD), IVTAN Thermodynamic Database, NIMS AtomWork, PoLyInfo (Polymer Database, NIMS), MatWeb, AZoM Materials Database, Crystallography Open Database (COD), PubChem, Springer Open Data (selected datasets), NASA Glenn Thermodynamic Database (NASA Polynomials).

## Interactive Presentation Requirements

The final platform should have good visual appeal and should allow the user to interact with the ($C_p$-T) curves. Desirable features include:

- Selection of a material from a searchable list or drop-down menu
- Selection of material categories
- Plotting of one or multiple materials simultaneously
- Comparison of ($C_p$) values of different materials
- User-defined temperature ranges
- Display of material name, formula, category, and data source
- Interactive zooming, panning, cursor values, or data tips
- Clear axis titles, units, legends, and labels
- Appropriate warning when the selected temperature lies outside the valid range of the thermodynamic equation
- Visually appealing graphs, dashboards, webpages, applications, or graphical user interfaces

Students are encouraged to include additional innovative features such as filtering, ranking, curve comparison, exportable graphs, phase-transition indicators, or material-property summaries.

**Submission Requirements**

**Important: Do Not Submit the Source Code**

The MATLAB or Python source code must **not** be submitted as the primary assignment submission.

Each group must submit:

1. A working link, executable application, interactive webpage, dashboard, or screen-recorded demonstration of the developed platform.
2. A short video demonstrating:
   - Selection of materials
   - Generation of $C_p$-T curves
   - Comparison of different materials
   - Interactive features of the platform
3. Two separate PowerPoint presentations: one prepared by each group member.

**Individual Presentation Requirement**

Although the computational platform will be developed jointly, each student must prepare and submit an individual PowerPoint presentation.

The two presentations from the same group must be clearly different in presentation style, organisation, visual design, explanation, and demonstrated skills.